

# 单位属性与卡池平衡设计文档

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## 设计目标

所有移植单位都进入卡池；开局完成地块类型和价格随机；单位/防御价格越高，高品质卡权重越高；上下地图以中心水平镜像，保证双方机会一致。

## 当前规则快照

| 地块             | 类型        | 价格模式        | 价格/池 | 产出 | 间隔  | 出现%  |
|----------------|-----------|-------------|------|----|-----|------|
| cell_question  | question  | fixed       | 25   | 0  | 0   | 50.0 |
| cell_unit      | unit      | random_pool | 0    | 0  | 0   | 20.0 |
| cell_defense   | defense   | random_pool | 0    | 0  | 0   | 20.0 |
| cell_gold_mine | gold_mine | fixed       | 50   | 10 | 3.0 | 10.0 |
| cell_home_base | home_base | free        | 0    | 12 | 3.0 | 0.0  |

## 价格池与品质分布

| 价格  | 概率%  | 品质档  | 单位卡池                 | 防御卡池                    |
|-----|------|------|----------------------|-------------------------|
| 50  | 30.0 | low  | unit_cards_price_50  | defense_cards_price_50  |
| 100 | 50.0 | mid  | unit_cards_price_100 | defense_cards_price_100 |
| 250 | 20.0 | high | unit_cards_price_250 | defense_cards_price_250 |

| 卡池                      | 条目 | 普通%  | 稀有%  | 史诗%  | 传说%  |
|-------------------------|----|------|------|------|------|
| defense_cards_price_100 | 4  | 25.0 | 40.0 | 25.0 | 10.0 |
| defense_cards_price_250 | 3  | 0    | 15.0 | 45.0 | 40.0 |
| defense_cards_price_50  | 3  | 70.0 | 25.0 | 5.0  | 0    |
| defense_cards_question  | 2  | 80.0 | 20.0 | 0    | 0    |
| unit_cards_price_100    | 18 | 29.0 | 46.0 | 20.0 | 5.0  |
| unit_cards_price_250    | 12 | 0    | 10.0 | 50.0 | 40.0 |
| unit_cards_price_50     | 12 | 65.0 | 30.0 | 5.0  | 0    |
| unit_cards_question     | 7  | 80.0 | 20.0 | 0    | 0    |

召唤单位全表

| 名称    | 族群          | 品质        | 费 | HP  | 速   | 射程  | 伤害/CD(DPS)     | 标签                         | 技能效果                                                |
|-------|-------------|-----------|---|-----|-----|-----|----------------|----------------------------|-----------------------------------------------------|
| 疾步狐   | beast       | common    | 2 | 85  | 5.0 | 1.1 | 10/0.9 (11.1)  | fast melee flanker         | 快速撕咬敌人并短暂加速 [damage haste; CD 0.9s]                 |
| 领嚎狼   | beast       | rare      | 4 | 135 | 4.2 | 1.2 | 18/1.1 (16.4)  | aura melee pack            | 嚎叫提升附近同族的攻击速度 [buff attack_speed pack; CD 7.0s]     |
| 雷掌熊   | beast       | epic      | 6 | 360 | 2.4 | 1.5 | 34/1.8 (18.9)  | tank stun high_hp          | 重击地面造成伤害和短暂眩晕 [damage stun shock; CD 6.0s]          |
| 冲牙奔袭者 | boarfolk    | common    | 3 | 145 | 4.1 | 1.2 | 16/1.2 (13.3)  | charge fast melee          | 向前冲锋并击退敌人 [damage dash knockback; CD 5.0s]          |
| 钢鬃守卫  | boarfolk    | rare      | 4 | 280 | 2.5 | 1.3 | 22/1.6 (13.8)  | tank retaliate             | 进入防御姿态并反刺攻击者 [armor retaliate thorns; CD 4.0s]      |
| 刺羽队长  | boarfolk    | epic      | 6 | 250 | 3.1 | 3.8 | 24/1.3 (18.5)  | ranged retaliate aura      | 齐射尖刺造成范围伤害 [damage projectile retaliate; CD 6.5s]   |
| 齿轮兵   | clockwork   | common    | 2 | 150 | 2.8 | 1.3 | 14/1.2 (11.7)  | sturdy melee               | 齿轮拳击造成近战伤害 [damage; CD 1.2s]                        |
| 炮台步机  | clockwork   | rare      | 5 | 180 | 2.3 | 5.4 | 32/1.9 (16.8)  | ranged high_attack slow    | 发射炮弹造成远程范围伤害 [damage aoe projectile; CD 5.5s]       |
| 修复核心  | clockwork   | epic      | 6 | 240 | 2.9 | 3.0 | 16/1.4 (11.4)  | support shield repair      | 展开修复场治疗并提供护盾 [heal shield repair; CD 8.0s]          |
| 火星幼龙  | dragonkin   | common    | 3 | 120 | 3.3 | 3.8 | 15/1.3 (11.5)  | ranged splash              | 喷吐火星造成小范围溅射 [damage splash burn; CD 1.3s]           |
| 天焰飞龙  | dragonkin   | epic      | 6 | 260 | 3.5 | 4.5 | 30/1.5 (20.0)  | ranged cone aoe            | 向前喷出扇形火焰 [damage burn cone; CD 7.5s]                |
| 古老巨龙  | dragonkin   | legendary | 9 | 520 | 2.6 | 5.2 | 42/2.0 (21.0)  | legendary ranged aoe       | 召下陨火造成大范围伤害 [damage burn stun; CD 11.0s]            |
| 电火元素  | elemental   | common    | 2 | 90  | 3.9 | 4.2 | 12/0.95 (12.6) | ranged chain               | 电弧弹射到额外目标 [damage chain; CD 0.95s]                  |
| 磐石元素  | elemental   | rare      | 4 | 310 | 2.0 | 1.2 | 20/1.7 (11.8)  | tank shield slow           | 生成岩壳护盾抵挡伤害 [shield armor; CD 9.0s]                  |
| 风暴元素  | elemental   | epic      | 6 | 210 | 3.4 | 4.8 | 26/1.2 (21.7)  | ranged chain burst         | 召唤链式风暴打击多个敌人 [damage chain shock; CD 6.5s]          |
| 烬爪小邪裔 | fiend       | common    | 2 | 70  | 4.7 | 1.2 | 13/1.0 (13.0)  | fast fragile burn          | 爪击附带持续灼烧 [damage burn; CD 1.0s]                     |
| 血契蛮徒  | fiend       | rare      | 4 | 165 | 3.3 | 1.4 | 27/1.4 (19.3)  | high_attack lifesteal risk | 牺牲少量生命换取吸血和狂暴 [lifesteal rage self_damage; CD 8.0s] |
| 裂隙领主  | fiend       | legendary | 8 | 430 | 2.7 | 2.0 | 48/1.9 (25.3)  | legendary summon burst     | 打开裂隙召唤援兵并造成爆发 [summon damage rift; CD 12.0s]        |
| 潮泽斥候  | marshkin    | common    | 2 | 100 | 4.4 | 1.5 | 11/0.9 (12.2)  | fast melee                 | 向前潮涌突刺并造成伤害 [damage dash; CD 4.5s]                  |
| 泥沼萨满  | marshkin    | rare      | 4 | 145 | 3.1 | 4.3 | 17/1.25 (13.6) | ranged poison regen        | 放置泥沼图腾让敌人中毒并回复友军 [poison regen zone; CD 8.0s]       |
| 泥浪斗士  | marshkin    | epic      | 6 | 300 | 2.8 | 1.5 | 29/1.5 (19.3)  | control tank slow          | 推出泥浪伤害并减速敌人 [slow damage wave; CD 7.0s]             |
| 弯刀海盗  | pirate      | common    | 3 | 125 | 3.5 | 1.3 | 18/1.05 (17.1) | melee high_attack          | 连续挥砍造成高频近战伤害 [damage combo; CD 1.05s]               |
| 火药炮手  | pirate      | rare      | 5 | 150 | 2.9 | 5.0 | 28/1.8 (15.6)  | ranged aoe bomb            | 投掷火药桶延迟爆炸 [damage aoe bomb; CD 6.5s]                |
| 舰队统领  | pirate      | legendary | 8 | 360 | 3.0 | 4.4 | 38/1.5 (25.3)  | legendary aura ranged      | 呼叫炮击并提升友军开火节奏 [damage aura barrage; CD 10.0s]       |
| 礁枪卫   | serpentfolk | common    | 3 | 115 | 3.6 | 3.6 | 16/1.2 (13.3)  | ranged pierce              | 投出穿透直线的礁枪 [damage pierce projectile; CD 1.2s]       |
| 鸣潮射手  | serpentfolk | rare      | 4 | 130 | 3.7 | 5.6 | 19/1.15 (16.5) | ranged focus kite          | 短暂专注以提高远程伤害 [buff damage focus; CD 6.0s]            |
| 潮汐先知  | serpentfolk | epic      | 6 | 210 | 3.2 | 4.8 | 20/1.35 (14.8) | support slow shield        | 为友军加盾并减速附近敌人 [shield slow water; CD 8.5s]           |
| 碎骨仆从  | undead      | common    | 2 | 75  | 3.2 | 1.1 | 9/1.0 (9.0)    | swarm fragile              | 唤起碎骨群涌向敌人 [summon swarm damage; CD 5.0s]            |
| 墓地射手  | undead      | rare      | 4 | 105 | 2.9 | 5.2 | 21/1.4 (15.0)  | ranged curse               | 射出诅咒箭降低目标防御 [damage curse projectile; CD 1.4s]      |
| 幽魂骑士  | undead      | epic      | 6 | 230 | 3.6 | 1.6 | 31/1.45 (21.4) | revive melee elite         | 首次倒下后化为幽魂短暂返场 [revive ghost haste; CD 12.0s]        |

防御塔全表

| 名称    | 品质        | HP  | 射程  | 伤害/CD(DPS)    | 标签                         | 技能效果                                          |
|-------|-----------|-----|-----|---------------|----------------------------|-----------------------------------------------|
| 哨戒箭塔  | common    | 180 | 5.2 | 16/1.2 (13.3) | tower ranged single_target | 射击最近的敌人并造成稳定伤害 [damage projectile; CD 1.2s]   |
| 火炮塔   | rare      | 240 | 5.8 | 34/1.9 (17.9) | tower ranged aoe           | 发射炮弹造成远程范围伤害 [damage aoe projectile; CD 5.5s] |
| 修复信标  | epic      | 220 | 3.5 | 0/2.0 (0.0)   | tower support repair       | 展开修复场治疗并提供护盾 [heal shield repair; CD 8.0s]    |
| 风暴方尖碑 | legendary | 300 | 6.0 | 42/2.2 (19.1) | tower legendary chain      | 召唤链式风暴打击多个敌人 [damage chain shock; CD 6.5s]    |

卡池明细

| 卡池                     | 卡牌    | 类型      | 品质     | 权重 | 概率%  |
|------------------------|-------|---------|--------|----|------|
| unit_cards_question    | 疾步狐   | unit    | common | 20 | 20.0 |
| unit_cards_question    | 烬爪小邪裔 | unit    | common | 15 | 15.0 |
| unit_cards_question    | 电火元素  | unit    | common | 15 | 15.0 |
| unit_cards_question    | 齿轮兵   | unit    | common | 15 | 15.0 |
| unit_cards_question    | 碎骨仆从  | unit    | common | 15 | 15.0 |
| unit_cards_question    | 领嚎狼   | unit    | rare   | 10 | 10.0 |
| unit_cards_question    | 炮台步机  | unit    | rare   | 10 | 10.0 |
| defense_cards_question | 哨戒箭塔  | defense | common | 80 | 80.0 |
| defense_cards_question | 火炮塔   | defense | rare   | 20 | 20.0 |
| unit_cards_price_50    | 疾步狐   | unit    | common | 16 | 16.0 |
| unit_cards_price_50    | 烬爪小邪裔 | unit    | common | 13 | 13.0 |
| unit_cards_price_50    | 电火元素  | unit    | common | 13 | 13.0 |
| unit_cards_price_50    | 齿轮兵   | unit    | common | 11 | 11.0 |
| unit_cards_price_50    | 潮泽斥候  | unit    | common | 12 | 12.0 |
| unit_cards_price_50    | 领嚎狼   | unit    | rare   | 8  | 8.0  |
| unit_cards_price_50    | 泥沼萨满  | unit    | rare   | 7  | 7.0  |
| unit_cards_price_50    | 鸣潮射手  | unit    | rare   | 5  | 5.0  |
| unit_cards_price_50    | 炮台步机  | unit    | rare   | 5  | 5.0  |
| unit_cards_price_50    | 磐石元素  | unit    | rare   | 5  | 5.0  |
| unit_cards_price_50    | 雷掌熊   | unit    | epic   | 3  | 3.0  |
| unit_cards_price_50    | 风暴元素  | unit    | epic   | 2  | 2.0  |
| unit_cards_price_100   | 火星幼龙  | unit    | common | 8  | 8.0  |
| unit_cards_price_100   | 弯刀海盗  | unit    | common | 8  | 8.0  |
| unit_cards_price_100   | 礁枪卫   | unit    | common | 7  | 7.0  |
| unit_cards_price_100   | 冲牙奔袭者 | unit    | common | 6  | 6.0  |

| 卡池                      | 卡牌    | 类型      | 品质        | 权重 | 概率%  |
|-------------------------|-------|---------|-----------|----|------|
| unit_cards_price_100    | 领嚎狼   | unit    | rare      | 8  | 8.0  |
| unit_cards_price_100    | 血契蛮徒  | unit    | rare      | 7  | 7.0  |
| unit_cards_price_100    | 炮台步机  | unit    | rare      | 7  | 7.0  |
| unit_cards_price_100    | 鸣潮射手  | unit    | rare      | 7  | 7.0  |
| unit_cards_price_100    | 墓地射手  | unit    | rare      | 7  | 7.0  |
| unit_cards_price_100    | 火药炮手  | unit    | rare      | 5  | 5.0  |
| unit_cards_price_100    | 钢鬃守卫  | unit    | rare      | 5  | 5.0  |
| unit_cards_price_100    | 雷掌熊   | unit    | epic      | 5  | 5.0  |
| unit_cards_price_100    | 天焰飞龙  | unit    | epic      | 5  | 5.0  |
| unit_cards_price_100    | 修复核心  | unit    | epic      | 5  | 5.0  |
| unit_cards_price_100    | 泥浪斗士  | unit    | epic      | 5  | 5.0  |
| unit_cards_price_100    | 舰队统领  | unit    | legendary | 2  | 2.0  |
| unit_cards_price_100    | 裂隙领主  | unit    | legendary | 1  | 1.0  |
| unit_cards_price_100    | 古老巨龙  | unit    | legendary | 2  | 2.0  |
| unit_cards_price_250    | 血契蛮徒  | unit    | rare      | 5  | 5.0  |
| unit_cards_price_250    | 炮台步机  | unit    | rare      | 5  | 5.0  |
| unit_cards_price_250    | 雷掌熊   | unit    | epic      | 9  | 9.0  |
| unit_cards_price_250    | 天焰飞龙  | unit    | epic      | 9  | 9.0  |
| unit_cards_price_250    | 风暴元素  | unit    | epic      | 8  | 8.0  |
| unit_cards_price_250    | 修复核心  | unit    | epic      | 7  | 7.0  |
| unit_cards_price_250    | 潮汐先知  | unit    | epic      | 5  | 5.0  |
| unit_cards_price_250    | 幽魂骑士  | unit    | epic      | 5  | 5.0  |
| unit_cards_price_250    | 刺羽队长  | unit    | epic      | 7  | 7.0  |
| unit_cards_price_250    | 裂隙领主  | unit    | legendary | 14 | 14.0 |
| unit_cards_price_250    | 古老巨龙  | unit    | legendary | 14 | 14.0 |
| unit_cards_price_250    | 舰队统领  | unit    | legendary | 12 | 12.0 |
| defense_cards_price_50  | 哨戒箭塔  | defense | common    | 70 | 70.0 |
| defense_cards_price_50  | 火炮塔   | defense | rare      | 25 | 25.0 |
| defense_cards_price_50  | 修复信标  | defense | epic      | 5  | 5.0  |
| defense_cards_price_100 | 哨戒箭塔  | defense | common    | 25 | 25.0 |
| defense_cards_price_100 | 火炮塔   | defense | rare      | 40 | 40.0 |
| defense_cards_price_100 | 修复信标  | defense | epic      | 25 | 25.0 |
| defense_cards_price_100 | 风暴方尖碑 | defense | legendary | 10 | 10.0 |
| defense_cards_price_250 | 火炮塔   | defense | rare      | 15 | 15.0 |

| 卡池                      | 卡牌    | 类型      | 品质        | 权重 | 概率%  |
|-------------------------|-------|---------|-----------|----|------|
| defense_cards_price_250 | 修复信标  | defense | epic      | 45 | 45.0 |
| defense_cards_price_250 | 风暴方尖碑 | defense | legendary | 40 | 40.0 |

## 平衡设计说明

50金币池强调铺场，普通卡占主要权重；100金币池承接中期阵容成形，稀有和史诗明显提高；250金币池作为高投入赌点，史诗和传说成为主体。防御池条目较少，目的是保持防御地块结果清晰，并方便后续按塔型扩展。

族群定位：野兽偏速度和群体攻速，邪裔偏高风险爆发，龙裔偏远程范围，元素偏弹射/护盾，机械偏炮击/修复，沼泽偏毒和回复，海蛇偏远程控制，海盗偏高攻和炮击，豪猪偏冲锋反击，亡灵偏群涌和返场。

外形资产状态：配置已经让所有移植单位进入卡池，但实际 sprite 资源仍需要按族群补齐。当前可以先通过颜色、缩放、武器/特效和程序化动作区分，后续再补专属序列帧。